

Natural History





About the Course

Learners dig deeper into relationships between creatures as they read about the beginnings of ecology. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 7

About Science: Grade 7

In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and beginning to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 7

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 7

In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and begin to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

Natural History: Grade 7

For approximately seventh-grade students, based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
7	Natural History: Grade 7 1 time/week 30 min	The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 7				
Natural History: Grade 7	General Science: Grade 7	General Science: Grade 7	Nature Notebook: Grade 7	Labs: Grade 7 Nature Walks & Scouting: Grades 1-8



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 7

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 7

Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

- ☐ Learn a few of your local species or varieties in connection with special topics.

Natural History: Grade 7

- ☐ Any term: natural history museum or butterfly house

Term Prep & Teacher Tips

Natural History: Grade 7

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - a flower with large, visible parts, such as a lily (Term 1)
 - a sunny window
 - water
 - a few sprigs of any water weed obtained from a pond or pet store that sells fish (Term 3)
 - various supplies by student design
 - optional: salt
 - optional: forceps/tweezers

Reminders

Natural History: Grade 7

- ☐ Note that the first lesson of Natural History is an activity, so be prepared with these and any other materials from the supply list on day 1.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Natural History: Grade 7



The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Natural History: Grade 7



Household Items - Natural History: Grade 7



Slides and Coverslips



Prepared Microscope Slides



Student Microscope



Quick Links

(No Quick Links Assigned)

[Click THIS text
or scan the QR
code for links.](#)



Natural History: Grade 7

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

- Notice if there were any dates that they want to keep for their Book of Centuries.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
- These can be viewed as alternative ways to engage.

Natural History: Grade 7

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 30m Natural History: Grade 7 - Lesson 1

Shoots Activity

Materials: 9 bean seeds, 3 recycled pots or cups, soil of any kind, aluminum foil (for later), sunny window, recycled box to hold 3 pots or cups, water

→ INTRO

This term, you are going to read the biography of a woman who began as a young artist and grew to be a skilled naturalist and biologist. We will begin as she did, studying the anatomy and physiology of flowering plants. Today, you will plant some bean seeds and observe how they "see" the light.

→ ACTIVITY

- Prepare 3 pots or cups with soil and plant 3 seeds in each cup.
- Set the cups near a sunny window, but inside a recycled box on its side, such that they only receive light from the direction of the window. Keep them watered so that the soil is damp, but not wet. They may only need 1-2 tablespoons of water at a time, depending on the size of your cups. It will take them 1-2 weeks to germinate, and you will need to keep the soil damp.

→ NARRATE

Draw or diagram your planted seed cups in their window in your science notebook.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 30m Natural History: Grade 7 - Lesson 2

The Girl in the Garden

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ INTRO

Let's begin reading about Maria Merian today. As we continue to learn about her and her work, we will see the interactions between plants, insects, and the environment. These shared relationships exist for all living things around the world and are called the study of ecology.

→ READ, NARRATE, & DISCUSS

"Prologue" p.viii-ix "A girl kneels" - "her story."

★ TEACHER TIP

The next 2 weeks are activity weeks!

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 30m Natural History: Grade 7 - Lesson 3

Shoots Activity

Materials: aluminum foil, water

→ INTRO

You should have shoots by now, and they are probably growing rapidly! Today, you will change their conditions to see if they respond.

→ ACTIVITY

- Cut 3 small squares of aluminum foil (about 2" x 2") and 3 small rectangles (about 1/2" x 3").
- Form the squares over your fingertip or a pencil to form 3 small caps.
- Wrap the rectangles around your finger or pencil to form 1/2" tall tubes.
- Place the caps on the tips of the seedlings in one cup; place the tubes around the bases of the seedlings in another cup. Leave the third cup alone - this is your control. Continue watering your cups this week.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 1

→ NARRATE

Draw or diagram your seedlings in your science notebook.

WEEK 4 30m Natural History: Grade 7 - Lesson 4

Shoots Activity

Materials: plantings from earlier in the term

→ INTRO

See how your shoots responded to their foil covers.

→ OBSERVE

Carefully observe how each of your seedlings grew this week. Are there differences? Does this reveal anything about how plants "see" light?

→ NARRATE

Draw or diagram your seedlings in your science notebook.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 30m Natural History: Grade 7 - Lesson 5

Egg

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.1 p.1-6 "Maria Sibylla" - "imagination instead?"

- Why do you think the author titled the first chapter "Egg"? Notice the chapter titles throughout the book.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 30m Natural History: Grade 7 - Lesson 6

Hatching

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.2 p.7-12 "Before Maria" - "about them."

★ TEACHER TIP

Are your learners demonstrating more familiarity/interest in plants? More interest/ease in discussing the relationships that plants have with animals and the rest of the environment? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 30m Natural History: Grade 7 - Lesson 7

First Instar

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

★ TEACHER TIP

Some students will benefit from reading and narrating the selection in parts because the two passages have different purposes. Can they identify the purpose of each

Natural History: Grade 7

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Term 1

→ RECAP

What did you read or observe last week?

→ READ & NARRATE

Ch.3 p.13-17 "Art was a joy" - "for profit."

→ READ, NARRATE, & DISCUSS

p.18-19 "Art was big" - "busy workshop."

→ STUDY

Maria had to study every detail of the plants she learned to paint. Study the diagram at the link and try to learn the parts of the leaf. Copy the diagram, if helpful.

∞ Image Link: Leaf Diagram

one?

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

WEEK 8 30m Natural History: Grade 7 - Lesson 8

Second Instar

📄 Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.4 p.20-22, 24-27 "Maria loved" - "her work."

→ READ, NARRATE, & DISCUSS

p.23 "Today" - "to record it."

★ TEACHER TIP

Note that students will need a flower with large, visible parts next week. A variety of lilies works well.

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

WEEK 9 30m Natural History: Grade 7 - Lesson 9

Flower Dissection Activity

📄 Materials: image, video, and article links, a flower with large, visible parts, optional: tweezers or toothpick

PREP: Read Teacher Tip

→ INTRO

Today, you will dissect a flower to examine all the parts carefully.

→ OBSERVE

Begin from the outside of the flower and work your way inward, carefully observing and removing each part. You can refer to the "Flower Anatomy" diagram at the link if desired.

∞ Image Link: Flower Anatomy

→ NARRATE

Draw or diagram your flower in your science notebook.

● Try to explain pollination using your diagram. On which part is the pollen? Where do the seeds or fruit develop?

→ SUPPLEMENTAL

You can learn how the seeds are fertilized and develop into fruit at the following links:

∞ Video Link: Germination

∞ Article Link: Seed and Fruit Development

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

Natural History: Grade 7

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Term 1

WEEK 10 30m Natural History: Grade 7 - Lesson 10

Third Instar

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.5 p.28-33 "Maria continued" - "was to marry."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 30m Natural History: Grade 7 - Lesson 11

Catch Up

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether plants or plant relationships in the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12 30m Natural History: Grade 7 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to the course.

Questions will come from:
The Girl Who Drew Butterflies

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 2

WEEK 1 30m Natural History: Grade 7 - Lesson 13

Fourth Instar

Materials: The Girl Who Drew Butterflies

→ RECAP

Recall where we left Maria Merian last term.

→ READ & NARRATE

Ch.6 p.34-38, 40 "At the age" - "for both."

→ READ, NARRATE, & DISCUSS

p.39 "Nearly 100,000" - "in society."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 30m Natural History: Grade 7 - Lesson 14

Molting

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.7 p.41-44 "Although tulip" - "to a branch."

→ READ, NARRATE, & DISCUSS

p.46-47 "As Maria discovered" - "fully understood."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 30m Natural History: Grade 7 - Lesson 15

Judge Not Too Quickly

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.7 p.48-54 "Sometimes" - "lot of effort."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4 30m Natural History: Grade 7 - Lesson 16

Pupa

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.8 p.55-60 "Encouraged" - "Waltha Castle."

• In this reading, we heard that Maria did not have a good relationship with her husband. Think about some healthy relationships that you have seen in your family or friends. Talk with a trusted adult or write about what makes a relationship healthy.

★ TEACHER NOTE

Maria Merian's estrangement from her husband is discussed on p.58-59. Teachers might choose to allow it to pass by the way or discuss it with students.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 2

WEEK 5 30m Natural History: Grade 7 - Lesson 17

Eclosing

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP

What did you read or observe last week?

→ READ & NARRATE

Ch.9 p.61-64, 66-67 "Windswept" - "scientific legacy."

→ READ, NARRATE, & DISCUSS

p.65 "In the 1600s" - "marriage."

- How should faith support our relationship with God, other people, and the world around us?

★ TEACHER NOTE

Some aspects of 17th-century religious practices/beliefs are mentioned on p.65, including the ordering of society, evolution, and marriage. Teachers should pre-read if they have concerns.

★ TEACHER TIP

Are your learners demonstrating more familiarity/interest in plants and ecosystem relationships? More interest/ease in discussing the relationships that plants have with the rest of the ecosystem? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 30m Natural History: Grade 7 - Lesson 18

Trouble

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.9 p.67, 70-73 "Spring came" - "to Amsterdam."

→ READ, NARRATE, & DISCUSS

p.68-69 "Maria called" - "for camouflage."

★ TEACHER TIP

The next 2 weeks are activity weeks!

★ TEACHER NOTE

The conclusion of Merian's unhappy marriage is presented on p.72: "Then trouble came" - "again a Merian." Teachers might choose to allow it to pass by the way or to engage in discussion.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 30m Natural History: Grade 7 - Lesson 19

Botanical Art Activity

Materials: The Girl Who Drew Butterflies, paper, paint, field guides, nature journal, or other records of observation

PREP: Read Teacher Tip

→ INTRO

Over the next 2 weeks, create botanical art in the style of Maria Merian using what you have learned about plant anatomy and its relationship with an insect. Use whatever

★ TEACHER TIP

Interested students may need additional time during their composition period this week and/or next to complete this in-class assignment.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 2

medium best suits you. You can even use a computer!

→ ACTIVITY

WEEK 8 30m Natural History: Grade 7 - Lesson 20

Botanical Art Activity

Materials: The Girl Who Drew Butterflies, paper, paint, field guides, nature journal, or other records of observation

→ ACTIVITY

Complete your botanical art.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 30m Natural History: Grade 7 - Lesson 21

Expanding

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.10 p.74-77 "After Maria's" - "or two."

→ READ, NARRATE, & DISCUSS

p.78-79 "Ever since" - "about evolution."

★ TEACHER NOTE

Evolution is mentioned toward the end of p.78. Teachers might choose to allow it to pass by the way or decide how they would like to discuss it.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 30m Natural History: Grade 7 - Lesson 22

Preparing

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.10 p.80-83 "Maria was" - "unknown land?"

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 30m Natural History: Grade 7 - Lesson 23

Catch Up

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether plants or the relationships that plants have with the rest of the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 2

WEEK 12  **30m Natural History: Grade 7 - Lesson 24**
Exam Week

→ **EXAMS**
Answer question(s) related to the course.

Questions will come from:
The Girl Who Drew Butterflies

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

Natural History: Grade 7

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Term 3

WEEK 1 30m Natural History: Grade 7 - Lesson 25

Flight

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.11 p.84-87 "In late" - "breeding caterpillars."

→ READ, NARRATE, & DISCUSS

p.88-89 "The history" - "in 1863."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 30m Natural History: Grade 7 - Lesson 26

New Lands

Materials: The Girl Who Drew Butterflies, video links

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.11 p.90-95 "Maria quickly" - "of sugarcane."

→ SUPPLEMENTAL

Scientists who study butterflies have been inspired in many ways:

∞ Video Link: Biomimicry and Butterflies

∞ Video Link: Butterfly-Inspired Robots

★ TEACHER NOTE

The experience of native and enslaved women, including suicide and the use of a plant to abort pregnancy, is mentioned on p.90-91: "The Indians" - "customs and lore." Teachers might choose to allow it to pass by the way, contributing to students' knowledge from history lessons, or engage students in discussion.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 30m Natural History: Grade 7 - Lesson 27

New Ideas

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.11 p.96-102 "Life was" - "had never seen."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4 30m Natural History: Grade 7 - Lesson 28

Egg

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.12 p.103-109 "By the time" - "on details."

• This chapter is titled "Egg," just like the first. What do you think is the egg that Maria left?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 3

WEEK 5 30m Natural History: Grade 7 - Lesson 29

Patience

Materials: The Girl Who Drew Butterflies

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

Ch.12 p.110-115 "Published in" - "miraculous lives."

- Discuss what 'ecology' is, based on Maria Merian's work.

You can hear or read more at the following links:

- ∞ Video Link: Introduction to Ecology
- ∞ Video Link: Why do we need insects?

★ TEACHER NOTE

The age of the Earth is mentioned in the second supplemental video (0:13). Teachers might choose to allow it to pass by the way, to discuss, or to skip this one.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 30m Natural History: Grade 7 - Lesson 30

In Her World

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP

What did you read or observe last week?

→ READ, NARRATE, & DISCUSS

p.116-121 "Maria Sibylla" - "it forever."

- "She saw nature as an ever-transforming web of connections..." What types of connections have you noticed in nature? Perhaps there will be some new ones in the video link:
- ∞ Video Link: Feedback Loops and Networks in Nature

★ TEACHER TIP

Are your learners demonstrating more familiarity/interest in ecological relationships? More interest/ease in discussing the life and work of Maria Merian? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• IMPORTANT DATES

Maria Merian (1647-1717)

WEEK 7 30m Natural History: Grade 7 - Lesson 31

An Ecological Story

Materials: video link

PREP: Read Teacher Tip

→ INTRO

Humans have an impact on the rest of the natural world. Because everything is ecologically connected, the full impact is often unexpected. Watch a short video during today's lesson.

→ VIEW & DISCUSS

- ∞ Video Link: The Hopeful Story of a Doomed Fox
- What do you see as the pros and cons of human intervention in this story? What is our responsibility when we disrupt an ecosystem?

★ TEACHER TIP

Next week is an activity week (as is the rest of the term). Students will need supplies, including a freshwater weed. This can be a bit of water weed from a pond (see this week's outdoor work) or any plant sold for freshwater fish tanks.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

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Term 3

WEEK 8 30m Natural History: Grade 7 - Lesson 32

Photosynthesis Activity

Materials: any freshwater weed, pH indicator, 4 tubes, holder for tubes such as a mug beaker or measuring cup, water, straw, aluminum foil, sunny window, microscope, slides and coverslips, methylene blue, table salt

→ INTRO

We return our focus to plants now, but with the deeper understanding of ecology that Maria Merian has taught us. One of the most amazing characteristics of plants is their ability to use the carbon dioxide gas that animals breathe out and, using the energy of sunlight, turn it into sugar and oxygen gas. This is called photosynthesis. Eventually, animals eat the plants and return the carbon to the atmosphere, completing the carbon cycle. Today, we examine chloroplasts, where photosynthesis happens in the plant, and even get to "see" them at work.

→ ACTIVITY

- ∞ Activity Link: Photosynthesis Activity
- ∞ Video Link: Wet Mount Slide of Elodea
- ∞ Video Link: Wet Mount with Salt Water

→ VIEW & NARRATE

At a convenient time later in the day, observe each of the 4 tubes. Do you notice any signs that the plant is using the carbon dioxide gas you added to the water? Do you notice any signs that the plant is releasing oxygen? Under what conditions? Draw or diagram your 4 tubes in your science notebook. If you want, you may return the tubes to their setup and observe one more time tomorrow. Why do you think it might be different tomorrow? Is there anything else you would like to investigate?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 30m Natural History: Grade 7 - Lesson 33

Composition Activity

Materials: by student design

→ INTRO

Over the next 2 weeks, create a composition of any type you like that describes the importance of plants for other creatures. You might refer to what you learned from Maria Merian or other books, or what you have learned in any of our activities this year. You might compose a poem, a piece of art, an essay, a slide presentation, etc.

→ ACTIVITY

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 30m Natural History: Grade 7 - Lesson 34

Composition Activity

Materials: by student design

→ ACTIVITY

Complete your composition activity from last week.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 30m Natural History: Grade 7 - Lesson 35

Catch Up

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether ecological relationships or Maria Merian are more familiar or have possibly even

Natural History: Grade 7

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Term 3	
→ CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? • FIELD TRIP Visit a natural history museum butterfly exhibit or butterfly house.
WEEK 12  30m Natural History: Grade 7 - Lesson 36 <i>Exam Week</i> → EXAMS Answer question(s) related to the course. Questions will come from: The Girl Who Drew Butterflies	• OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Examination

- Term 1** · The Girl Who Drew Butterflies: Draw a diagram illustrating how plant anatomy is dependent on animal life and environmental conditions.
- Term 2** · The Girl Who Drew Butterflies: Explain how close observation revealed new ideas to Maria that others did not see.
- Term 3** · The Girl Who Drew Butterflies: Explain how animals are ecologically dependent on plants. OR Give reasons why Maria Merian is sometimes called the first ecologist.